

**Examining Reliance Patterns on AI Advice in Medical Imaging: A Mixed-Methods
Randomized Crossover Experiment**

Authors

Susanne Gaube (PhD)^{1,2*}, Ekaterina Jussupow (PhD)³, Eesha Kokje (PhD)², Timo K. Koch
(PhD)^{2,4}, Jowaria Khan (BSc)⁵, Elizabeth Bondi-Kelly (PhD)⁵, Andreas Schicho (MD)⁶, Felipe C.
Kitamura (PhD)^{7,8}, Timur Ezer (state exam)⁹, Juergen Mottok (PhD)⁹, Eva Lerner (Prof, PhD)^{2,10},
Marzyeh Ghassemi (PhD)¹¹, Errol Colak (MD)^{13,14}

Affiliations

¹UCL Global Business School for Health, University College London, London, UK

²Department of Psychology, LMU Munich, Munich, Germany

³Department of Law and Economics, Technical University of Darmstadt, Darmstadt, Germany

⁴Institute of Behavioral Science and Technology, University of St. Gallen, St. Gallen, Switzerland

⁵Department of Computer Science and Engineering, University of Michigan, Ann Arbor, USA

⁶Department of Radiology, University Hospital Regensburg, Regensburg, Germany

⁷Eden, Palo Alto, USA

⁸Departamento de Diagnóstico por Imagem, Universidade Federal de São Paulo (Unifesp), São
Paulo, Brazil

⁹Faculty of Electrical Engineering and Information Technology, Ostbayerische Technische
Hochschule, Regensburg, Germany

¹⁰Department of Business Psychology, Technical University of Applied Sciences Augsburg,
Augsburg, Germany

¹¹Electrical Engineering and Computer Science, Institute for Medical Engineering and Science;
Massachusetts Institute of Technology, Cambridge, USA

¹³Department of Medical Imaging, St. Michael's Hospital, Unity Health Toronto, Toronto, Canada

¹⁴Department of Medical Imaging, Faculty of Medicine, University of Toronto, Toronto, Canada

27 Correspondence concerning this article should be addressed to Susanne Gaube, UCL East –
28 Marshgate, 7 Sidings St, London E20 2AE, UK. E-Mail: susanne.gaube@ucl.ac.uk

Research in Context

Evidence before this study

Artificial intelligence (AI) is transforming healthcare. However, its integration into clinical workflows does not necessarily improve clinical outcomes. Our study builds on the seminal 2022 review by Rajpurkar et al., which summarized the landscape of AI in medicine and called for more research into human-AI collaboration. We searched Google Scholar multiple times using search terms such as “reliance” AND “artificial intelligence” AND (“clinical decision-making” OR “medical imaging”), along with relevant synonyms, to identify collaboration paradigms in clinical settings. While many clinical studies have reported on the risks and causes of overreliance, i.e., users following incorrect AI advice, few have discussed underreliance, i.e., users dismissing correct AI advice, as a key error mode. To our knowledge, no study has comprehensively examined how underreliance compares to overreliance in its impact on diagnostic performance and the underlying reasons.

Added value of this study

Our study systematically investigates and contrasts overreliance and underreliance in AI-supported diagnostic decision-making. Using a multi-session crossover experiment, we exposed participants to AI advice of varying accuracy and explainability during CT scan review. We combined high case volume and varying case complexity with controlled advice accuracy, while also deploying think-aloud and eye-tracking methods to uncover mechanisms behind error modes. We show that underreliance harms diagnostic performance more than overreliance in absolute terms. Moreover, while explainable AI (XAI) improved performance and reduced uncertainty-driven errors, it did not mitigate underreliance rooted in rejecting the system or use errors.

Implications of all the available evidence

Current efforts to improve human-AI collaboration focus heavily on reducing overreliance through explainability. However, our findings suggest that underreliance also poses a significant risk to achieving human-AI synergy. As AI models become more accurate, the risk of dismissing correct AI advice will increase in both frequency and consequence. AI development should move beyond

Abstract

Background: Artificial intelligence (AI) holds significant potential to support diagnostic decision-making; however, evidence regarding its clinical utility remains mixed. Often, the collaboration between clinicians and AI systems does not surpass the individual performance of unaided humans or standalone AI. Yet, currently, the mechanisms that limit human-AI synergy are poorly understood. This study examined the impact of AI advice on diagnostic decision-making among experts and novices, focusing on reliance patterns.

Methods: We used a mixed-methods crossover experimental design with a think-aloud and an eye-tracking study arm. Participants were 50 task experts (radiologists) and 75 novices (non-radiologist physicians and medical trainees) from 10 countries. They reviewed 50 head CT scans and every case was examined in three time-separate sessions in randomized order. In each session, participants were exposed to different experimental conditions: (a) *control*, no AI prediction; (b) *basic advice*, AI prediction without annotations; and (c) *XAI advice*, AI prediction with scan annotations. For each case, participants had to determine if the patients had an intracranial hemorrhage (ICH). The main outcomes were diagnostic performance, confidence in the diagnosis, case reading time, and AI advice usefulness ratings.

Findings: Both overreliance on incorrect advice and underreliance on correct advice occurred. Underreliance was associated with high uncertainty and, in absolute terms, had a more detrimental impact on diagnostic performance than overreliance. Correct XAI advice reduced underreliance, improved performance ($OR=1.84$, $p<0.0001$), and confidence ($b=0.15$, $p<0.0001$), particularly when reviewing more difficult cases with ICH. Surprisingly, correct XAI did not reduce reading time ($b=1.81$, $p=0.0713$). XAI was perceived as more useful than basic AI advice ($b=0.12$, $p=0.0029$), especially among novices.

Interpretation: The occurrence of both under- and overreliance highlights the need for efficient counterstrategies beyond classic XAI methods to foster appropriate reliance and synergy.

Funding: Volkswagen Foundation, Forum for Interdisciplinary Research, German Federal Ministry

Introduction

Artificial intelligence (AI) is rapidly transforming healthcare, impacting diagnostics, prognostics, treatment, and drug discovery.¹ Many AI models perform at or above the level of clinicians in various clinical tasks.¹ However, randomized controlled trials report mixed results, with many showing minimal and no significant effects on decision-making, diagnostic performance, and patient outcomes.² Moreover, many AI models have limited generalizability, i.e., they perform well in their development environment but deteriorate in external settings.³

AI models are often implemented as Clinical Decision Support Systems (AI-CDSS), where clinicians can incorporate the model's results into their decisions.⁴ Consequently, the effectiveness of an AI-CDSS depends on its users. Ideally, collaboration between clinicians and AI systems should outperform unaided human and standalone AI performance, creating *human-AI synergy*.^{1,5} While synergy is rare across use domains,⁵ AI support often improves human performance (i.e., *human augmentation*).⁵

The most frequently cited reason for the failure to achieve human-AI synergy is *overreliance*, where users follow incorrect AI advice, a phenomenon well-documented in clinical tasks^{6–13} and in the wider human factors literature.¹⁴ Explanations for overreliance include automation bias, discounting of AI, anchoring, confirmation bias, high trust in technology/beliefs in AI capabilities, and cognitive offloading.^{7,9,11,15} The second reason for human-AI underperformance is *underreliance*, where users fail to follow correct AI advice.^{5,8,15–17} Compared to overreliance, underreliance has received less attention, both in user studies and the AI implementation discourse, with only a few clinical studies even reporting this error pattern.^{11,12,17,18} Although regulators have explicitly called for measures to mitigate overreliance,¹⁹ the impact of users dismissing correct AI advice seems underappreciated. This is particularly concerning as AI models become increasingly more accurate.¹ Some factors contributing to underreliance have been proposed, including algorithm aversion, lack of trust, perceived AI inaccuracy, or AI-specific limitations.^{11,20} However, empirical evidence contrasting the impact of underreliance with

121 negative effects.^{8,10,11,13} However, methodological differences may account for the mixed findings.
122 First, several studies involved small sample sizes,^{8,11,12,18,22,23} and/or tested very few
123 cases.^{6,8,10,11,22–24} Second, some studies used sequential paradigms,^{8,12} where participants first
124 make an initial decision, then view AI advice, and then make a final decision; others use
125 concurrent paradigms, where participants make a single decision after reviewing the case and AI
126 advice.^{6,10,13,16,24} Although both designs offer insights into decision-making, clinical workflows
127 typically involve reviewing both the case and AI advice before completing a report, as delayed AI
128 input would compromise efficiency.⁷ More empirical evidence is needed from XAI user studies with
129 robust and diverse samples, high case volumes, and varying case complexity, while mimicking
130 clinical workflows in a controlled environment. This will help determine if, when, and how XAI can
131 enhance human-AI collaboration.

132 Our study analyzes the impact of both overreliance and underreliance on AI advice in diagnostic
133 decision-making using a randomized controlled crossover experiment among 50 experts and 75
134 novices. Participants reviewed 50 cases with varying difficulty, containing head CT scans and
135 patient information, under three conditions: no advice, concurrent basic AI advice, and concurrent
136 XAI advice (Figure 1). By manipulating the accuracy of AI advice, we examined whether
137 explainability helps avoid overreliance on incorrect and underreliance on correct advice (Figure 1d
138 defines reliance patterns). Moreover, some participants partook in a think-aloud or an eye-tracking
139 study arm to gain a better understanding of the underlying mechanisms. Our results offer valuable
140 insights into optimizing human-AI collaboration in healthcare.

Methods

Participants

Guided by our hypotheses (supplements pp3-4), simulation-based power analysis (supplements p5),²⁵ and preregistration (supplements p6), participants with differing task expertise were recruited: *experts* (general radiologists, neuroradiologists, and radiology residents from year two onwards) and *novices* (medical students, interns, and non-radiologist physicians). All gave written informed consent after being briefed on the study's purpose and procedure. The study received ethics approval (# 21-2475_2-101). Recruitment and compensation are described in the supplements (p5).

Experimental Design

We conducted a paired crossover experiment where participants reviewed 50 cases containing head CT scans and patient information (task rationale in supplements p7). Each participant reviewed every case in three sessions in a randomized order with a minimum 14-day washout period between sessions. Participants were randomly assigned to a study group. In each session, participants were exposed to a different condition: *control*, *basic advice*, and *XAI advice*. In the *control* condition, they reviewed cases without support. In the other conditions, participants received diagnostic support labeled as a report from an AI-CDSS, which the authors generated. This was done to avoid confounding effects caused by case complexity and prevalence of disorders, as recommended in the literature.²⁶ The AI report provided either *basic advice*, consisting only of the AI prediction, or *explainable advice* (XAI), including the prediction and visible annotations (Figure 1). The *accuracy* of the advice was manipulated, with 80% being correct. The 80-20 ratio was chosen because it corresponds with the lower end of AI accuracy rates in clinical settings,²⁷ and ensures sufficient statistical power to assess the impact of incorrect advice. In the XAI condition, prediction and annotation accuracy aligned (i.e., correct predictions were paired with correct annotations and vice versa). The selection of cases with inaccurate advice was

171 and 50% without ICH (negative cases; ICH-), with 10 easy, 20 medium, and 20 hard cases, evenly
172 distributed. Three versions were created for each case: not annotated, correctly annotated, and
173 incorrectly annotated. We chose bounding boxes as annotations because they are among the
174 most widely used localization-based XAI methods in FDA-cleared AI products for medical image
175 interpretation.^{1,28} The selection, difficulty assessment criteria, and annotation processes are
176 further described in the supplements (pp8-10). Case IDs, patient information, difficulty ratings, and
177 ground truths are listed in the supplements (p11).

178 **Procedure**

179 In the first session, participants completed a survey that included demographics and other control
180 measures (supplements p15). In each session, they had to review all 50 CTs and determine if a
181 patient had an ICH. In the advice conditions, participants were informed they could incorporate the
182 AI's prediction into their decision-making. In both advice conditions, the AI prediction (ICH
183 detected: yes/no) was displayed in the interface's top row (supplements p10). The localization-
184 based XAI advice condition featured side-by-side CTs, with and without annotations. Following the
185 diagnosis, participants rated their confidence in the decision and, in advice conditions, the AI
186 advice's usefulness before proceeding to the next case. After completing all cases, participants
187 evaluated their experience and received instructions for the next session. After the washout
188 period, they were invited to the second session, which included the experiment and session
189 evaluation. This procedure was repeated for the final session. Further procedural details are in the
190 supplements (p14).

191 **Outcome Measures**

192 **Performance:** Diagnostic performance was assessed by comparing participants' decisions ("Does
193 this patient have intracranial hemorrhage (ICH): *yes/no*") with the ground truth, yielding a binary
194 variable: 0 for inaccurate and 1 for accurate decisions.

195 **Confidence:** Participants' confidence in their diagnosis was measured using one question ("How
196 confident are you with your decision?") rated on a Likert scale from 1 (*not at all*) to 5 (*extremely*).

197 **Reading time:** The individual case reading time was measured in seconds from starting a case to
198 clicking the 'next' button for the subsequent case.

199 **Usefulness:** The AI advice's perceived usefulness was assessed with one question ("How useful
200 was the AI report for your decision?") rated on a Likert scale from 1 (*not at all*) to 5 (*extremely*).

201 **Data Analysis**

202 Mixed-effects regression models were fitted for each outcome. Performance was analyzed using
203 logistic mixed-effects regression. Linear mixed-effects regression was used for confidence ratings,
204 reading times, and advice usefulness ratings. By design, the experiment produced a rank-deficient
205 fixed-effect model matrix because the control condition has no advice. Therefore, for the
206 performance, confidence, and reading time outcomes, condition and advice accuracy were
207 combined into a five-level factor: no advice, correct basic advice, incorrect basic advice, correct
208 XAI advice, and incorrect XAI advice. These outcomes were regressed on the combined factor,
209 expertise, their interaction, and control variables. Advice usefulness was analyzed in the advice-
210 only subset regressing on advice format, advice accuracy, expertise, interactions, and control
211 variables. Participants and cases were included as random intercepts to account for the non-
212 independence of observations, participants' skills, and case difficulty. Further information on the
213 data analysis is available in the supplements (pp16-18)

214 **Mixed Methods**

215 All participants completed the web-based experiment, and a subset was non-randomly assigned
216 to an additional study arm, either think-aloud (novices) or eye-tracking (experts), to gain deeper
217 insights into their decision-making. Methodological details are in the supplements (pp19-23).

218 **Role of the Funding Source**

219 The funders had no role in the study design, data processing, manuscript preparation, or
220 submission decision.

Results

Of the 163 eligible participants who signed up, 125 started the first session and were included in the data analysis. 111 completed all three sessions, four completed two, and five completed one full session. The mean age was 28.48 years ($SD = 6.72$), and 55% (69/125) self-identified as female. Participants came from 10 countries. The 50 task experts had an average of 7.85 years of professional experience ($SD = 5.85$), while the 75 novices had 0.17 years ($SD = 1.50$). Additional demographics are reported in the supplements (p24).

Underreliance Impacted Performance for More Patient Cases

The rate of overreliance ranged from 27% (111/417) to 58% (421/730) across expertise levels and AI conditions (supplements p25). It was higher than the underreliance rate, which ranged from 11% (186/1664) to 21% (597/2880). However, since 80% of the advice was correct, underreliance affected more patient cases (Figure 2 and supplements p25). If considering the base-rate, the net-impact of underreliance on diagnostic performance (i.e., the improvement in accuracy if underreliance is omitted minus the improvement in accuracy if overreliance is omitted) ranges from 5% for experts with basic advice, 4% for experts with XAI advice, to 5% for novices with basic advice and -1% for novices with XAI advice. Consequently, underreliance, in absolute terms, impacted performance more negatively than overreliance (supplements pp26-27). Underreliance was more pronounced among positive (ICH+) and more difficult cases (supplements p37). Importantly, XAI strongly reduced underreliance on true positive cases (i.e., ground truth ICH+ and AI report ICH+) and only slightly increased overreliance on false positive cases (i.e., ground truth ICH- and AI report ICH+; see Figure 2 and supplements p25).

Patterns Causing Underreliance

From the think-aloud data (supplements pp56-59), we identified four error patterns for underreliance among novices (supplements p60). First and foremost, novices showed *medical uncertainty* and often could not verify the advice or the annotation as their assessment

243 *uncertainty* and often could not verify the advice or the annotation, as their assessment
246 contradicted it. For example, if the AI indicated no bleeding and participants found suspicious
247 areas, but expressed uncertainty about their assessment. Or, the basic AI correctly indicated
248 bleed, but participants could not find it. Second, novices *rejected* the AI advice because they did
249 not trust it, suspected specific system errors, or found alternative explanations for the AI advice.
250 Third, novices *ignored* the AI and did not include its assessment in their decision-making, often
251 due to experiencing suspected errors in previous interactions. When rejecting and ignoring the AI,

252 participants were highly confident. Lastly, only in true negative cases (i.e., ground truth ICH- and
253 AI report ICH-), novices agreed with the no-bleeding advice but *accidentally clicked* bleeding. The
254 distribution of error patterns differs across conditions (supplements p60). Receiving correct XAI
255 advice strongly reduced errors caused by medical uncertainty. However, XAI did not mitigate
256 errors due to actively rejecting the AI.

257 From the eye-tracking data, we identified error patterns among experts, complementing the think-
258 aloud findings from novices. We measured the number of fixations, defined as points where
259 participants' gaze remained relatively stable within pre-defined areas of interest. Interestingly,
260 the number of fixations in the area containing the bleeding was highest when experts rejected
261 correct XAI advice in true positive cases (Figure 2f). Simultaneously, the number of fixations in
262 other areas of the CT was high, indicating that underreliance on XAI was associated with
263 uncertainty, but led to actively rejecting the AI advice, since participants clearly focused on the
264 proposed area. In contrast, when experts rejected correct basic advice in true positive cases, the
265 number of fixations in the area containing the bleeding was very low, while the number of fixations
266 on other regions of the CT was extremely high (Figure 2e), indicating that underreliance on basic
267 advice was also associated with uncertainty, but the error was caused by missing the bleeding.
268 When participants missed a bleeding (ground truth ICH+), no fixations in the area containing the
269 bleeding were recorded in 38% (17/45) of cases with correct and 67% (18/27) with incorrect basic
270 advice. These numbers were drastically reduced to 3% (1/37) for correct XAI while remaining at
271 similar levels 65% (13/20) for the incorrect XAI cases, suggesting that correct XAI prevents
272 participants from ignoring the advice and overlooking the suggested bleeding.

273 **XAI Advice Improved Diagnostic Performance**

274 Novices' baseline performance at 63% (2288/3650) [61% - 64%] was significantly lower than
275 experts' at 83% (1824/2200) [81% - 84%] (Table 1; means in supplements p28). Receiving correct
276 advice, both basic and XAI, increased performance; however, experts only significantly benefited
277 from XAI (Figure 3a). Even when receiving correct advice, novices only outperformed the AI

285 performance in true positive cases, with novices even outperforming experts when both groups
286 received XAI advice for difficult cases. When advice was incorrect, XAI negatively influenced
287 novices in easier, false positive cases (Figure 3f) but had no significant effect otherwise (Figure
288 3c-e).

289 **XAI Advice Increased Confidence in Diagnostic Decisions**

290 Novices' baseline confidence at $M = 3.09$ [3.05-3.12] was significantly lower than experts' at $M =$
291 4.14 [4.10-4.18] (Table 1; means in supplements p29). Correct XAI advice increased confidence,
292 while correct basic advice did not. Incorrect advice, whether basic or XAI, decreased confidence.
293 The effects of AI advice on confidence were more pronounced among novices (Figure 4,
294 supplements pp42-43).

295 While performance gains and losses from receiving advice were intuitively plausible, the picture is
296 more complex when considering participants' confidence (Figure 4a and b). Among both groups,
297 while performance was higher for negative cases (true negative and false positive), confidence
298 ratings tended in the opposite direction. Experts gained confidence from receiving XAI compared
299 to basic advice in true positive and, surprisingly, true negative cases, while novices only gained
300 confidence from XAI in true positive cases.

301 Compared to their baselines, experts and novices were less confident in their decisions when
302 underrelying on correct advice. Experts were more confident when receiving XAI compared to
303 basic advice, both when showing appropriate reliance and overreliance (Figure 4c). Novices
304 reported higher confidence when receiving XAI compared to basic advice when they appropriately
305 relied upon advice, but also when they showed underreliance (Figure 4d).

306 **XAI Advice Did Not Reduce Reading Time**

307 Novices' baseline reading time at $M = 52.35$ [51.43 - 53.27] was not significantly different from
308 experts at $M = 53.59$ [51.80 - 55.38] (Table 1; means in supplements p30). Receiving correct

309 basic advice decreased reading time overall; however, this effect was driven by experts and was
310 not observed among novices, who showed longer reading times in true positive cases (Figure 4,
311 supplements pp46-47). Overall, correct XAI advice did not significantly reduce reading time; a
312 reduction was observed only among experts in true-negative cases (Figure 4e). Incorrect advice,
313 whether XAI or basic, generally increased reading time, except in false negative cases (i.e.,
314 ground truth ICH+ and AI report ICH-), particularly among novices (Figure 4f).

315 Reading times were higher when underrelying on correct advice than at baseline (Figure 4g and
316 h). Novices also took longer to review cases when appropriately relying on themselves in cases of
317 incorrect advice. Among experts and novices, reading time was significantly higher for XAI advice
318 than basic advice when they appropriately relied on correct advice. Reading time was also higher
319 among novices overrelying on incorrect XAI advice.

320 **Participants Found XAI Advice More Useful**

321 Participants rated XAI and correct advice as more useful than basic and incorrect advice
322 (supplements pp31,35). Receiving XAI advice was more useful for novices than for experts, but
323 novices also regarded inaccurate advice as more useful than experts, indicating that they were
324 less capable of distinguishing between correct and incorrect advice (supplements p50). XAI
325 advice was consistently judged more useful than basic advice when reviewing true positive cases
326 (supplements pp51,53,54). Incorrect advice, especially false negative advice, was judged less
327 useful when reviewing easier than more difficult cases (supplements pp52,54).

328 **Discussion**

329 This study examined reliance patterns on basic and explainable (XAI) AI advice in clinical
330 decision-making. While many AI systems perform at or above clinician level,¹ their integration into
331 practice does not necessarily improve clinical practice.² Our findings indicate that underreliance on
332 correct AI advice may contribute to this, as it harmed diagnostic performance more than
333 overreliance on incorrect advice in absolute terms. While overreliance has been extensively
334 studied,^{6–13} this is one of the first studies to systematically explore the impact of and reasons for
335 underreliance in a clinical context. While the impact depends on the prevalence of correct AI
336 advice, our contribution lies in quantifying the conditional reliance patterns that determine whether
337 improvements in AI accuracy translate into performance gains.

338 Underreliance was associated with high uncertainty, leading to strong engagement with the CT
339 scans, resulting in the highest case reading time and the lowest confidence ratings. Expanding

346 from uncertainty. Aligned with this finding, XAI was most beneficial for novices, especially in more
347 difficult cases, confirming previous research.²⁹ Due to their stronger reliance on AI advice, novices
348 even outperformed experts when receiving XAI advice for more difficult, true positive cases.
349 However, only correct basic advice reduced overall case reading time, while incorrect advice,
350 whether basic or XAI, increased reading time. This suggests that the additional information XAI
351 provides requires more processing time but results in better performance. Finally, XAI was
352 perceived as more useful, especially in true positive cases. However, novices also regarded XAI
353 as more useful in false positive cases and generally when overrelying. In summary, our findings
354 may help clarify mixed results in previous clinical XAI user studies,^{6,8,10,12,13,16,22–24,29} by highlighting
355 how explainability interacts with case type and complexity, as well as user experience, suggesting
356 that one-size-fits-all XAI solutions are suboptimal.³⁰

357 Given that the accuracy of AI models improves rapidly, the distribution between correct and
358 incorrect advice will become increasingly skewed. Consequently, future research should focus
359 more on understanding underreliance using longitudinal designs to assess how reliance patterns
360 evolve with increasing familiarity and experience with AI. Moreover, to optimize the use of AI-
361 CDSS in clinical settings, we need to look for optimal and customizable XAI methods that support
362 users in relying on correct AI advice while being able to detect and overrule incorrect advice.
363 Finally, given that several instances of underreliance occurred due to errors of use (clicking the
364 wrong button) and not errors of judgment, we discourage the implementation of interfaces that
365 only require clinicians to agree or disagree with AI-generated advice and highlight the importance
366 of usability testing.

367 The current study has limitations. Most importantly, we prioritized experimental control by using
368 artificially generated advice with a fixed accuracy level within a web-based interface and by
369 employing a single imaging task with a binary decision outcome rather than a full report. Although
370 this design strengthens internal validity and enables isolation of reliance patterns, it creates an
371 artificial setting that may limit ecological validity. In real-world clinical environments, AI systems
372 exhibit greater variability and error patterns, decision-making carries tangible risk, medical

373 decisions are inherently uncertain, and task characteristics differ across specialties. Replication in
374 real-world settings and across other clinical contexts is necessary to establish generalizability of
375 our findings. Next, the distribution of easy, medium, and hard cases, as well as the distribution
376 between ICH+ and ICH- cases, was not derived from a realistic setting. We oversampled hard and
377 positive cases to enable a well-powered systematic investigation of reliance patterns across cases
378 of differing complexity. Third, despite the two-week washout period and randomization of case and

379 advice presentations, there remains the potential for recall bias, particularly for difficult or unusual
380 cases. Fourth, the expert group included early-career radiologists who might be more prone to rely
381 on AI advice. And finally, participation in the think-aloud and eye-tracking study may have
382 influenced behavior by making participants feel more observed, but decision patterns remained
383 consistent across all groups.

384 **Conclusion**

385 This study highlights that we need to advance our understanding of why clinicians underrely on
386 correct AI advice rather than focusing only on overreliance in clinical decision-making. While XAI
387 can reduce underreliance, it can also increase processing time. Future research should develop
388 novel approaches that bolster reliance on correct AI advice while improving efficiency and
389 enabling clinicians to identify and override errors to improve human-AI synergy.
390

Acknowledgment

We extend our gratitude to everyone who assisted in testing the experimental platform, as well as the radiologists and students who piloted our study. Special thanks to our (student) research assistants, Leander Kahle, Stefan Rogge, and Marco Weiß, for their efforts in collecting, transcribing, and cleaning the think-aloud data, and to Anna Heinrich for organizing the eye-tracking data collection. We also thank Vaibhav Balloli for contributing to the initial code for pre-processing the eye-tracking data. We want to express our gratitude to the FHInvest program "Eyes on Future" and Prof. Dr. Juergen Mottok, along with his eye-tracking team, consisting of Florian Hauser, Lisa Grabinger, and Timur Ezer, for procuring and providing the eye-tracking hardware and software used in this project. We also extend our thanks for the continuous maintenance of this laboratory and its equipment. Furthermore, we are grateful to Prof. Dr. Christian Wolff for providing facilities at the University of Regensburg and setting up the laboratory. Finally, we thank Thomas Weiß and Thomas Herbst for developing the user interface and experimental platform. Participants were compensated for their contributions.

Conflict of Interest Disclosures

Susanne Gaube reported receiving corporate sponsorship in the form of an unrestricted gift to support excellent research in academia from Google Ireland Limited, unrelated to the submitted work. Ekaterina Jussupow reported receiving funding from the DFG and Hessian.AI for scientific work on medical decision-making building on this paper. Additionally, Ekaterina Jussupow received fees for presentations and panel participations not related to this project. Eesha Kokje reported receiving funding from the LMU mentoring program unrelated the submitted work. Jowaria Khan was supported by the University of Michigan. Elizabeth Bondi-Kelly was partly funded by the University of Michigan, Google, and Microsoft. She is also the President of a nonprofit corporation called Try AI, Inc. Andreas Schicho received payments for expert group advisory from Bayer Healthcare. Felipe C. Kitamura reported personal fees as a consultant for Eden, Bunkerhill Health, GE Healthcare, and MD.ai, as well as serving as a speaker for Sharing Progress in Cancer Care outside the submitted work. He also received fees to participate in the

417 Progress in Cancer Care outside the submitted work. He also received fees to participate in the
418 Data Safety Monitoring Board for the LuANA Trial (Hospital Israelita Albert Einstein). He also
419 disclosed unpaid roles as Early Career Consultant to the Editor of Radiology, Associate Editor of
420 Radiology: Artificial Intelligence, Vice-chair of the SIIM ML Committee, and a member of the
421 RSNA AI Committee and the RSNA Radiology Informatics Council outside the submitted work.
422 Eva Lerner reported serving as the project lead for the PSY-AI-EYE project, which is funded by
423 the Bavarian State Ministry for Science and the Arts. Additionally, Eva Lerner receives book

royalties from Hogrefe, UTB, Reinhardt, and Springer for authored books and book chapters. Marzyeh Ghassemi was partly funded by the Hermann L. F. von Helmholtz Career Development Professorship at MIT, Microsoft Research, a CIFAR Global Scholar, and an NSF CAREER Award. She served as an unpaid board member for the AHLI non-profit, and as an unpaid co-chair of a CHAI working group. Errol Colak would like to acknowledge the support received from the Odette Professorship in Artificial Intelligence for Medical Imaging, St Michael's Hospital, Unity Health Toronto. He reports an unpaid role as Chair of the RSNA Artificial Intelligence Committee. No other disclosures were reported.

Funding/Support

This work was supported by a grant from the Volkswagen Foundation (Project "Human-AI-Interaction in Healthcare: Identifying Factors Contributing to Clinical Utility"; Grant #: 98525, 98526, 98527). Parts of the think-aloud participant compensation and data cleaning were paid for by the Forum for Interdisciplinary Research: Project "Metacognition in Human-AI Interaction" and the "Starter-Kit" by the Vereinigung von Freunden der Technischen Universität zu Darmstadt e.V.. The eye-tracking equipment was funded by the FHInvest program "Eyes on Future" (Grant #: 13FH101IN6) of the German Federal Ministry of Education and Research.

Data Sharing Statement

Data collected for this study are openly available. No participant identifiers are included. The dataset includes: (1) de-identified participant-level data from the main experiment; (2) de-identified eye-tracking data; (3) de-identified think-aloud transcripts of overreliance patterns and coded data; (4) R scripts used for statistical analyses of the main experiment; and (5) R scripts used for analysis of the eye-tracking data. All materials are available via the project's Open Science Framework (OSF) repository at: <https://doi.org/10.17605/OSF.IO/DQRA6>. The data are publicly accessible and will remain available without an end date. Additional related documents are available via the same repository, including study preregistrations. Data are shared without restriction and may be used for any lawful research purpose without investigator approval. We

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the authors used ChatGPT and Microsoft Copilot to improve readability and language. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Author Contributions

S.G.: Conceptualization, Methodology, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing – review & editing, Validation, Visualization, Project administration, Responsible for the submission decision

E.J.: Methodology, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing – review & editing, Funding acquisition

E.K.: Methodology, Formal analysis, Data curation, Writing - original draft, Writing – review & editing, Visualization

T.K.K.: Methodology, Software, Validation, Writing - review & editing

J.K.: Software, Data curation

E. B.-K.: Resources, Data curation, Writing - review & editing

A.S.: Conceptualization, Validation, Resources, Writing - review & editing, Funding acquisition

F.C.K.: Validation, Resources, Writing - review & editing

T.E.: Resources, Data curation

J.M.: Resources

E.L.: Supervision, Funding acquisition

M.G.: Conceptualization, Writing - review & editing, Funding acquisition

E.C.: Conceptualization, Validation, Writing - review & editing, Supervision, Funding acquisition

All authors had access to the raw data.

References

- 476
- 477 1 Rajpurkar P, Chen E, Banerjee O, Topol EJ. AI in health and medicine. *Nat Med* 2022; **28**: 31–
478 8.
- 479 2 Han R, Acosta JN, Shakeri Z, Ioannidis JPA, Topol EJ, Rajpurkar P. Randomised controlled
480 trials evaluating artificial intelligence in clinical practice: a scoping review. *Lancet Digit Health*
481 2024; **6**: e367–73.
- 482 3 Futoma J, Simons M, Panch T, Doshi-Velez F, Celi LA. The myth of generalisability in clinical
483 research and machine learning in health care. *Lancet Digit Health* 2020; **2**: e489–92.
- 484 4 Hummelsberger P, Koch TK, Rauh S, *et al*. Insights on the current state and future outlook of AI
485 in health care: Expert interview study. *JMIR AI* 2023; **2**: e47353.
- 486 5 Vaccaro M, Almaatouq A, Malone T. When combinations of humans and AI are useful: A
487 systematic review and meta-analysis. *Nat Hum Behav* 2024; **8**: 2293–303.
- 488 6 Jabbour S, Fouhey D, Shepard S, *et al*. Measuring the impact of AI in the diagnosis of
489 hospitalized patients: A randomized clinical vignette survey study. *JAMA* 2023; **330**: 2275.
- 490 7 Jussupow E, Spohrer K, Heinzl A, Gawlitza J. Augmenting medical diagnosis decisions? An
491 investigation into physicians' decision-making process with artificial intelligence. *Inf Syst Res*
492 2021; **32**: 713–35.
- 493 8 Du Y, Antoniadis AM, McNestry C, McAuliffe FM, Mooney C. The role of XAI in advice-taking
494 from a clinical decision support system: A comparative user study of feature contribution-based
495 and example-based explanations. *Appl Sci* 2022; **12**: 10323.
- 496 9 Gaube S, Suresh H, Raue M, *et al*. Do as AI say: Susceptibility in deployment of clinical
497 decision-aids. *Npj Digit Med* 2021; **4**. DOI:10.1038/s41746-021-00385-9.
- 498 10 Jacobs M, Pradier MF, McCoy TH, Perlis RH, Doshi-Velez F, Gajos KZ. How machine-learning
499 recommendations influence clinician treatment selections: the example of antidepressant
500 selection. *Transl Psychiatry* 2021; **11**: 1–9.
- 501 11 Bussone A, Stumpf S, O'Sullivan D. The role of explanations on trust and reliance in clinical
502 decision support systems. In: Proceedings of the 2015 International Conference on Healthcare
503 Informatics. Dallas, TX, USA: IEEE, 2015. DOI:10.1109/ICHI.2015.5.
- 504 12 Cao S, Liu A, Huang C-M. Designing for appropriate reliance: The roles of AI uncertainty
505 presentation, initial user decision, and user demographics in AI-assisted decision-making. *Proc*

- 514 15Vasconcelos H, Jörke M, Grunde-McLaughlin M, Gerstenberg T, Bernstein MS, Krishna R.
515 Explanations can reduce overreliance on AI systems during decision-making. *Proc ACM Hum-*
516 *Comput Interact* 2023; **7**: 1–38.
- 517 16Chanda T, Hauser K, Hobelsberger S, *et al.* Dermatologist-like explainable AI enhances trust
518 and confidence in diagnosing melanoma. *Nat Commun* 2024; **15**: 524.
- 519 17Wies C, Hauser K, Brinker TJ. Reply to: False conflict and false confirmation errors are crucial
520 components of AI accuracy in medical decision making. *Nat Commun* 2024; **15**: 6897.
- 521 18Calisto FM, Santiago C, Nunes N, Nascimento JC. BreastScreening-AI: Evaluating medical
522 intelligent agents for human-AI interactions. *Artif Intell Med* 2022; **127**: 102285.
- 523 19European Parliament and the Council of the European Union. Artificial Intelligence Act. 2024
524 <http://data.europa.eu/eli/reg/2024/1689/oj/eng> (accessed March 7, 2025).
- 525 20Jussupow E, Benbasat I, Heinzl A. An integrative perspective on algorithm aversion and
526 appreciation in decision-making. *MIS Q* 2024; **48**: 1575–90.
- 527 21Hassija V, Chamola V, Mahapatra A, *et al.* Interpreting black-box models: A review on
528 explainable artificial intelligence. *Cogn Comput* 2024; **16**: 45–74.
- 529 22Hwang J, Lee T, Lee H, Byun S. A clinical decision support system for sleep staging tasks with
530 explanations from artificial intelligence: User-centered design and evaluation study. *J Med*
531 *Internet Res* 2022; **24**: e28659.
- 532 23Sivaraman V, Bukowski LA, Levin J, Kahn JM, Perer A. Ignore, trust, or negotiate:
533 Understanding clinician acceptance of AI-based treatment recommendations in health care. In:
534 Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems. Hamburg
535 Germany: ACM, 2023: 1–18.
- 536 24Calisto FM, Fernandes J, Morais M, *et al.* Assertiveness-based agent communication for a
537 personalized medicine on medical imaging diagnosis. In: Proceedings of the 2023 CHI
538 Conference on Human Factors in Computing Systems. Hamburg Germany: ACM, 2023: 1–20.
- 539 25Pargent F, Koch TK, Kleine A-K, Lerner E, Gaube S. A tutorial on tailored simulation-based
540 sample-size planning for experimental designs with generalized linear mixed models. *Adv*
541 *Methods Pract Psychol Sci* 2024; **7**: 25152459241287132.
- 542 26Yu F, Moehring A, Banerjee O, Salz T, Agarwal N, Rajpurkar P. Heterogeneity and predictors of
543 the effects of AI assistance on radiologists. *Nat Med* 2024; **30**: 837–49.

- 544 27 Agarwal S, Wood D, Grzeda M, *et al.* Systematic review of artificial intelligence for abnormality
545 detection in high-volume neuroimaging and subgroup meta-analysis for intracranial hemorrhage
546 detection. *Clin Neuroradiol* 2023; **33**: 943–56.
- 547 28 McNamara SL, Yi PH, Lotter W. The clinician-AI interface: intended use and explainability in
548 FDA-cleared AI devices for medical image interpretation. *Npj Digit Med* 2024; **7**: 80.
- 549 29 Gaube S, Suresh H, Raue M, *et al.* Non-task expert physicians benefit from correct explainable
550 AI advice when reviewing X-rays. *Sci Rep* 2023; **13**: 1383.

551 30Famiglini L, Campagner A, Barandas M, La Maida GA, Gallazzi E, Cabitza F. Evidence-based
552 XAI: An empirical approach to design more effective and explainable decision support systems.
553 *Comput Biol Med* 2024; **170**: 108042.

554

